

Marcus Sostak

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Active Secret Clearance

EDUCATION

University of South Carolina

Columbia, SC

Bachelor of Science in Computer Science, Minor in Mathematics | **GPA: 3.92**

May 2028

- **Relevant Coursework:** Software Engineering, Data Structures & Algorithms, Advanced Programming (C++), Operating Systems, Computer Architecture
- **Organizations:** Vice President of Gamecock AI/ML Club, Infrastructure Chair of Kappa Theta Pi Professional Technical Fraternity

EXPERIENCE

Research Intern

May 2026 – August 2026

Applied Research Laboratory for Intelligence and Security, University of Maryland

College Park, MD

- Designed an AI-aware Enterprise Governance Framework, integrating AI/ML considerations into decision rights, data governance, and risk management across 8+ domains.
- Built an AI use portfolio of 10+ assessed cases, mapping real operational friction points to AI-assisted workflows.
- Developed a standards-aligned AI evaluation schema to score 3 government models on task performance, robustness, data/OPSEC risk, prompt-injection susceptibility, and bias/harm.
- Authored Responsible AI Adoption Recommendations and an implementation roadmap for agency leadership, detailing technical and policy guidance for scaling AI/ML capabilities securely.

Undergraduate Research Assistant

May 2025 – May 2026

University of South Carolina

Columbia, SC

- Developed and maintained an educational web platform used by local elementary schools for interactive word-based learning using RedwoodJS, React, GraphQL APIs and PostgreSQL.
- Refactored legacy frontend and backend components, removing 1,000+ lines of redundant code and improving maintainability, stability, and deployment reliability.
- Implemented automated preview environments integrated with CI, reducing code review turnaround time by ~ 50% and decreasing post-deployment bugs.
- Collaborated with a faculty mentor and a team of 4 developers on a \$250K+ grant-funded research project, using Trello to plan Agile sprints, prioritize features, and ship user-driven improvements.

PROJECTS

Utility Pole Inspection AI — 1st Place GridStorm Hackathon | *Python, YOLO, PaddleOCR, Godot, Android*

- Built an end-to-end inspection platform using a Godot Android client to send field images to a FastAPI backend that orchestrates YOLOv26 and PaddleOCR to generate structured JSON asset records.
- Designed a unified inference API combining OCR, object detection, and segmentation in a single request, maintaining <1 second average latency per inspection.
- Reduced cold-start latency by 65% via async model preloading and constrained inference to smaller model sizes with checkpoint ensembling to balance accuracy and sub-second response time.

Dialect Classifier App | *React, FastAPI, Docker, TailwindCSS, JavaScript, Supabase*

- Deployed a full-stack web application with a Dockerized backend on Render and a React frontend on Vercel, enabling production-ready data handling for real-time dialect classification.
- Coded FastAPI backend services for audio capture, metadata persistence, and RESTful APIs using Supabase, supporting collection of 1000+ labeled training samples while implementing clear user consent disclosures.
- Designed a modular, scalable backend architecture to support future audio preprocessing pipelines and ML-backed inference services.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL, JavaScript, TypeScript, MATLAB

Frameworks & Libraries: React, Flask, RedwoodJS, FastAPI

AI/ML: PyTorch, pandas, NumPy, PaddleOCR, Ultralytics YOLO

Developer Tools: Git, GitHub, Linux, Docker, AWS (S3), Render, Vercel

Databases & APIs: PostgreSQL, MySQL, Supabase, GraphQL, REST